DAY 2: SQL CODING CHALLENGE – DML:

ONLINE BOOKSTORE 📚

Question 1: CREATE TABLE with PRIMARY & FOREIGN KEY

Scenario:

You are creating a database for an online bookstore.

Task:

- Create a table Books with columns:
  - o BookID → INTEGER, PRIMARY KEY
  - o Title → VARCHAR(100), NOT NULL
  - o Author → VARCHAR(50), NOT NULL
  - o ISBN → VARCHAR(20), UNIQUE
  - o Price → DECIMAL(8,2), CHECK(Price > 0)
- Create a table Orders with columns:
  - o OrderID → INTEGER, PRIMARY KEY
  - o BookID → INTEGER, FOREIGN KEY REFERENCES Books(BookID)
  - o OrderDate → DATE, NOT NULL
  - o Quantity → INTEGER, CHECK(Quantity > 0)

Expected Output:

Tables are created successfully with all constraints applied.

Question 2: ALTER TABLE – Add UNIQUE Constraint

Scenario:

The bookstore wants to make sure ISBN is unique for every book.

Task:

- Alter the Books table to add a UNIQUE constraint to the ISBN column.

Expected Output:

The ISBN column enforces uniqueness.

Question 3: INSERT, RETRIEVE & UPDATE with Constraints

Scenario:

You want to add sample book data and update certain records.

Task:

- Insert at least 5 records into the Books table, respecting all constraints.
- Retrieve all records to verify entries.
- Update the Price or Quantity for a specific record while maintaining the CHECK constraints.

Expected Output:

All entries and updates comply with constraints and are displayed correctly.

Question 4: DELETE vs TRUNCATE

Scenario:

The bookstore wants to manage orders by removing some rows or clearing all data.

Task:

- Use DELETE with a WHERE clause to remove specific rows from Orders table.
- Use TRUNCATE to remove all rows while keeping table structure intact.

Expected Output:

- DELETE removes selected rows.
- TRUNCATE clears all rows quickly but keeps the table structure.